1. Given an array of characters formed with 'a' and 'b'. The string is marked with a special character 'X' representing the middle of the list.
   Example: ababaXababa
   Check whether the string is palindrome or not using stack.

2. Given a stack, reverse the elements of the stack using only stack operations (push and pop).
3. How do you check whether the stack grows up (toward higher memory addresses) or down (toward lower memory addresses)?

4. Let Q be a nonempty queue and let S be an empty stack. Using only the abstract operations for the queue and stack ADTs, write a program to reverse the order of items in Q.

5. Implement a Deque (doubly ended queue), where elements can be added or removed from both the front (head) or back (tail). It is also called a head-tail linked list.

6. Given a stack of integers, check whether each successive pair of numbers in the stack is consecutive or not. The pairs can be increasing or decreasing, if the stack has an odd number of elements, the element at the top is left out of a pair. For example, if the stack of elements are [4, 5, -2, -3, 11, 10, 5, 6, 20], then the output should be true because each of the pairs (4, 5), (-2, -3), (11, 10), and (5, 6) consists of consecutive numbers.

7. Given an integer k and a queue of integers, reverse the order of the first k elements of the queue, while keeping the other elements in the same relative order.
   Example:
   - Input: k=4, queue = [10, 20, 30, 40, 50, 60, 70, 80, 90];
   - Output: [40, 30, 20, 10, 50, 60, 70, 80, 90].

8. Given a queue of integers, rearrange the elements by interleaving the first half with the second half.
   Example:
   - Input: [11, 12, 13, 14, 15, 16, 17, 18, 19, 20].
   - First half = [11, 12, 13, 14, 15]
   - second half: [16, 17, 18, 19, 20].
   - Output: [11, 16, 12, 17, 13, 18, 14, 19, 15, 20].